

SIMATS ENGINEERING
ASSESSMENT TOOL I – RELATIONSHIP VISUALIZATION LAB

Course: Data Handling and Visualization	Code: DSA0603	CO: CO4
Duration: 60 min	Total Marks: 50	Reg No : 192524225 Name:VALLAPU MOHAN KUMAR

COURSE OUTCOME COVERED

CO1 Understand the fundamental concepts, principles, and techniques of data visualization. (BL3) Activity Tool : Relationship Visualization Lab
Weightage: 20%

Code of Conduct:

I VALLAPU MOHAN KUMAR (192524225) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V. Mohan Kumar

Criteria	Selection of Relationship Visualization 25%	Relationship Analysis 25%	Application of Visualization Techniques 20%	Organization 15%	Accuracy 10%	Clarity 5%	Total
Q1							

Q2							
Q3							
Q4							
Q5							

Relationship Visualization Lab

Provided Datasets

The following files are supplied alongside this document so the lab can be completed without internet access to install extra data packages:

- `mtcars.csv` — real R built-in dataset (32 cars $\times$ 11 variables: `mpg`, `cyl`, `disp`, `hp`, `drat`, `wt`, `qsec`, `vs`, `am`, `gear`, `carb`)
- `iris.csv` — real R built-in dataset (150 flowers $\times$ 5 variables: `Sepal.Length`, `Sepal.Width`, `Petal.Length`, `Petal.Width`, `Species`)
- `gapminder2007_practice.csv` — illustrative practice dataset in the style of the Gapminder 2007 cross section (country, continent, year, `gdpPercap`, `lifeExp`, `pop`). Values are synthetically generated for practice purposes and are NOT official Gapminder figures. If internet access is available, students may instead run `install.packages("gapminder")` and use `data(gapminder)` for the authentic dataset.
- `network_edges.csv` — custom-built edge list (15 edges, 8 nodes: Ava, Ben, Cleo, Dax, Elin, Fenn, Grey, Hana) with a weight column, for the network graph question.

Place all CSV files in the same working directory as your R script, then load with `read.csv("filename.csv")`.

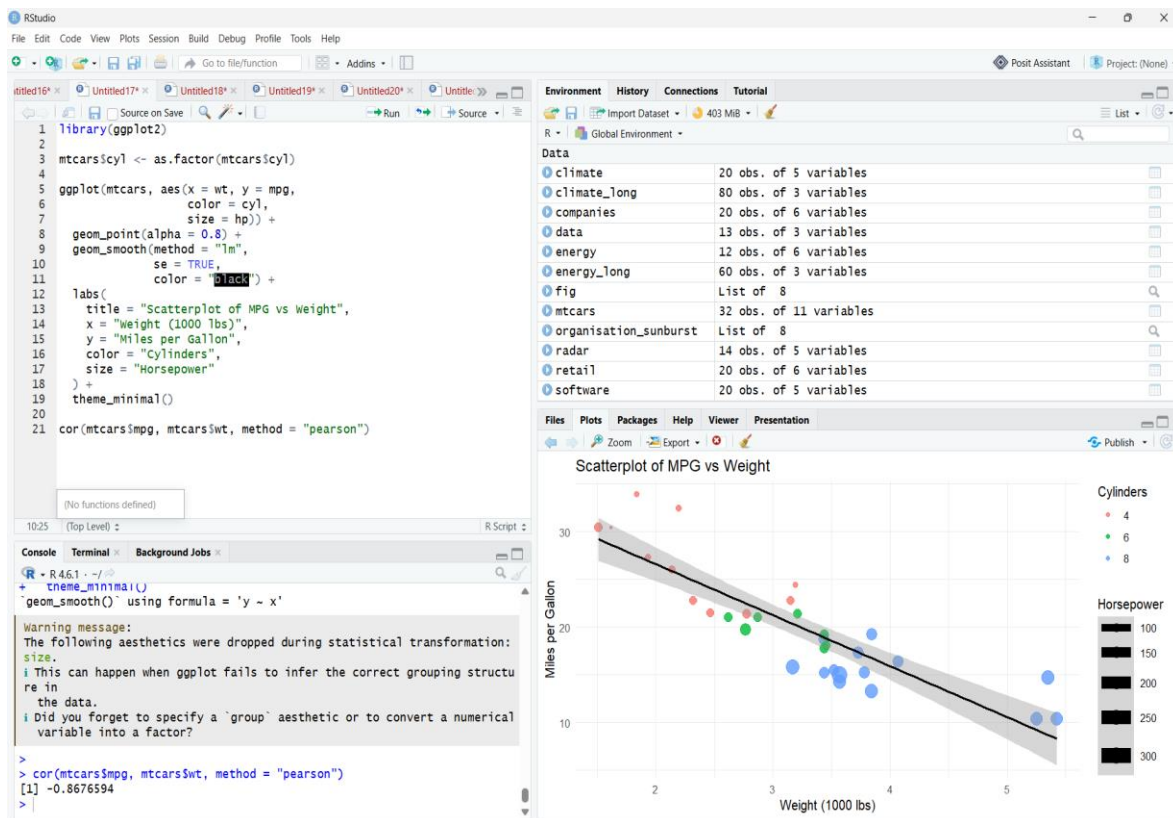
Question 1: Scatterplot with Regression Diagnostics

Dataset provided: `mtcars.csv`

Using the `mtcars` dataset, create a scatterplot of `mpg` vs `wt` (weight) using `ggplot2`. Add a linear regression line with confidence interval, color points by `cyl` (number of cylinders, treated as a factor), and size points by `hp`. Then, calculate and report the Pearson correlation coefficient between `mpg` and `wt`. Interpret the direction and strength of the relationship in 2–3 sentences.

Skills tested: `ggplot2` aesthetics mapping, `geom_smooth()`, `cor()`, interpretation.

Answer:



There is a **strong negative relationship** between car weight and mileage. As the weight of a car increases, its miles per gallon (mpg) generally decreases. The correlation value of approximately **-0.868** indicates a strong inverse relationship.

Graph

The scatterplot should show:

- wt on the X-axis
- mpg on the Y-axis
- Different colors for cylinder groups
- Larger points for higher horsepower
- A downward-sloping regression line
- A confidence interval around the regression line

Question 2: Correlogram Construction

Dataset provided: mtcars.csv (all numeric columns)

Using the mtcars dataset, compute the correlation matrix and visualize it as a correlogram using either the corplot or GGally::ggcorr package. Configure the plot to:

- Display correlation coefficients as numeric labels
- Use a diverging color scale (blue–white–red)
- Reorder variables using hierarchical clustering (order = "hclust" if using corplot)

Identify the two variable pairs with the strongest positive and strongest negative

correlations. **Skills tested:** Correlation matrices, corplot/GGally, hierarchical reordering, reading correlograms.

Answer:



Variable Pair	Correlation
cyl – disp	0.902
disp – hp	0.791
mpg – wt	-0.868

mpg – disp	-0.848
------------	--------

Strongest Positive Correlations

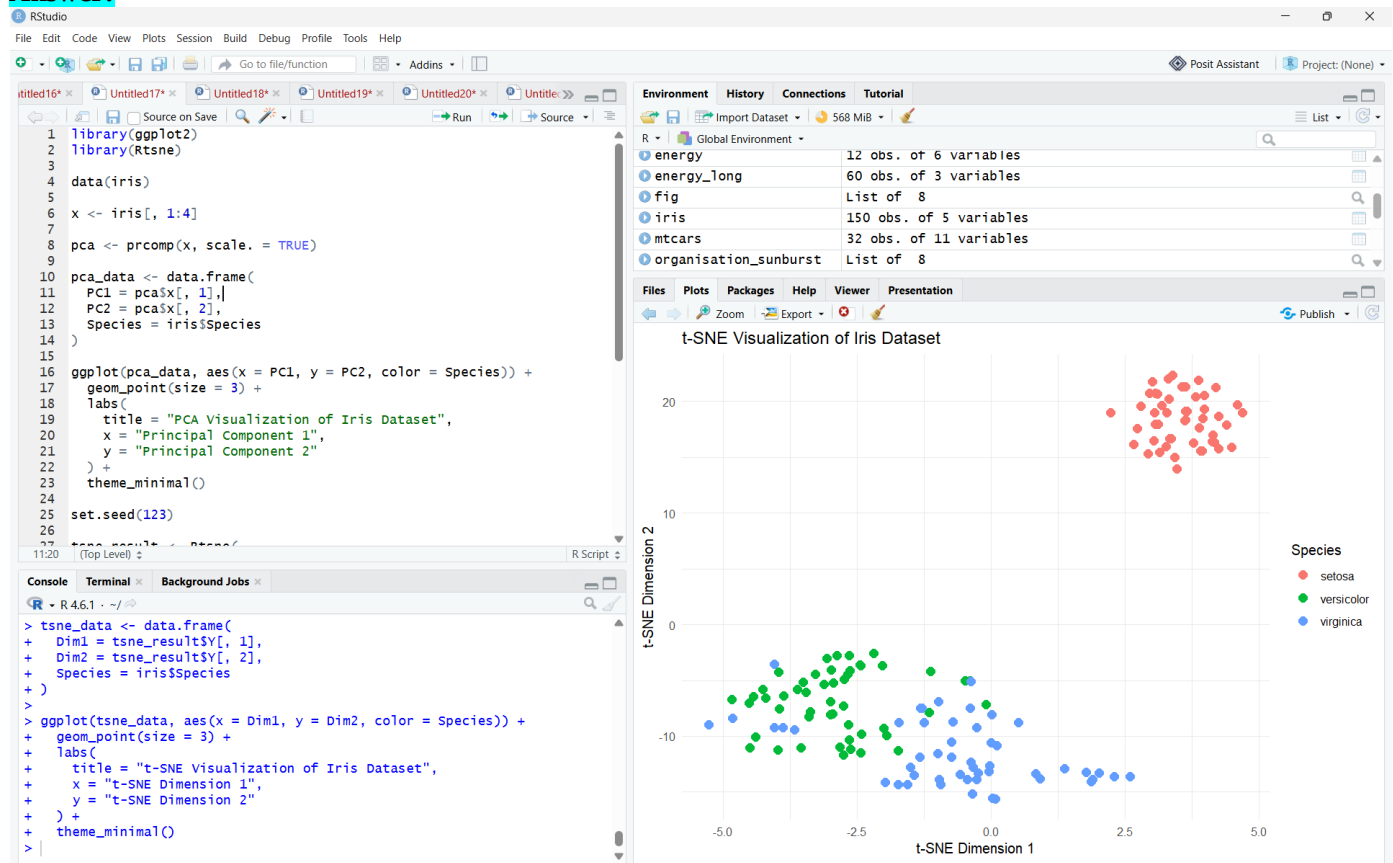
1. cyl and disp = 0.902
2. disp and hp = 0.791

Strongest Negative Correlations

1. mpg and wt = -0.868
2. mpg and disp = -0.848

Question 3: Dimension Reduction Comparison (PCA vs. t-SNE)

Answer:



Feature	PCA	t-SNE
Type	Linear	Non-linear

Main purpose	Variance preservation	Local similarity visualization
Speed	Faster	Relatively slower
Interpretation	Easier	More difficult
Dimensionality	Usually 2 or 3 components	Usually 2 or 3 dimensions
Best use	Overall structure	Cluster visualization

Question 4: Bubble Chart for Multivariate Paired Data

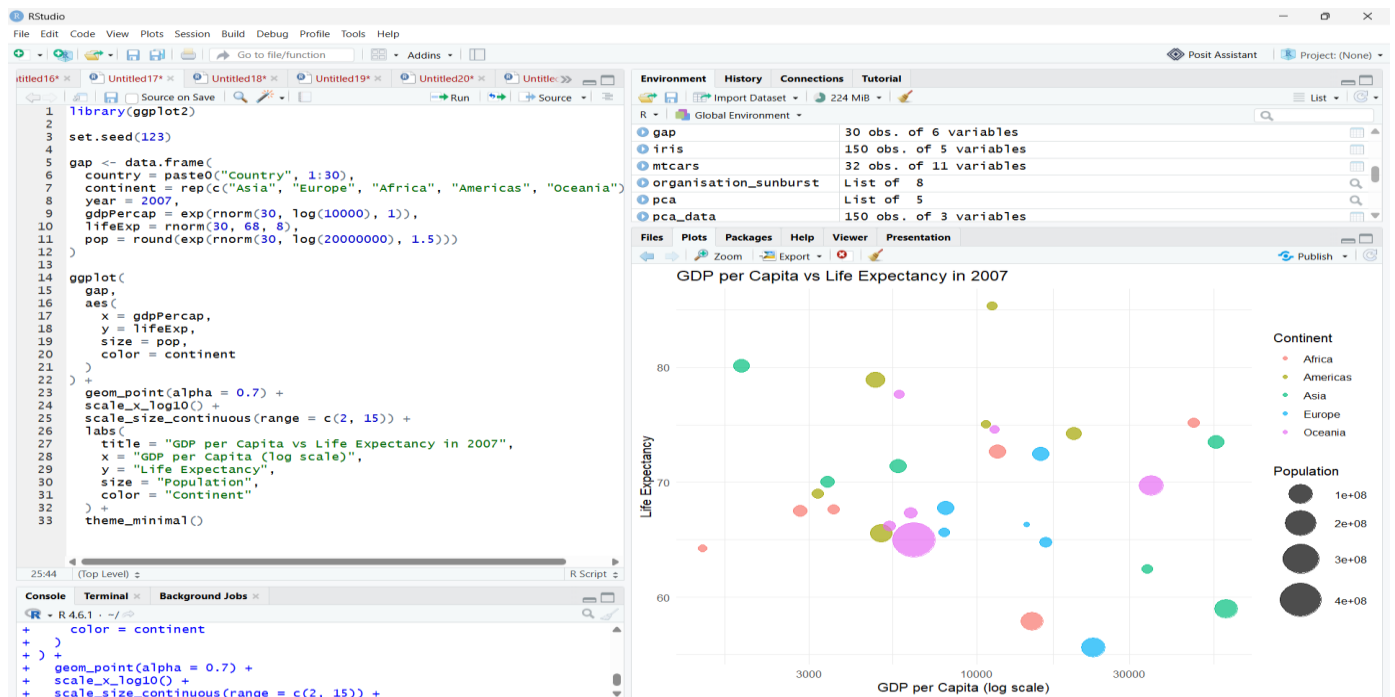
Dataset provided: gapminder2007_practice.csv (or the gapminder package, year == 2007) Filter for the year 2007 and create a bubble chart showing:

- x-axis: gdpPercap (log-transformed scale)
- y-axis: lifeExp
- bubble size: pop
- bubble color: continent

Add appropriate axis labels, a title, and use `scale_size_continuous()` to control bubble size range so points don't overplot excessively. Discuss one visible cluster or outlier in the data.

Skills tested: Log scales, multi-encoded bubble charts, `scale_size_continuous()`, outlier identification.

Answer:



What you should get

You will get a **bubble chart** with:

- **X-axis:** GDP per capita, log scale
- **Y-axis:** Life expectancy
- **Bubble size:** Population
- **Bubble color:** Continent

Question 5: Hexbin Plot vs. Network Graph

This question has two parts to compare visualization strategies for different relationship types.

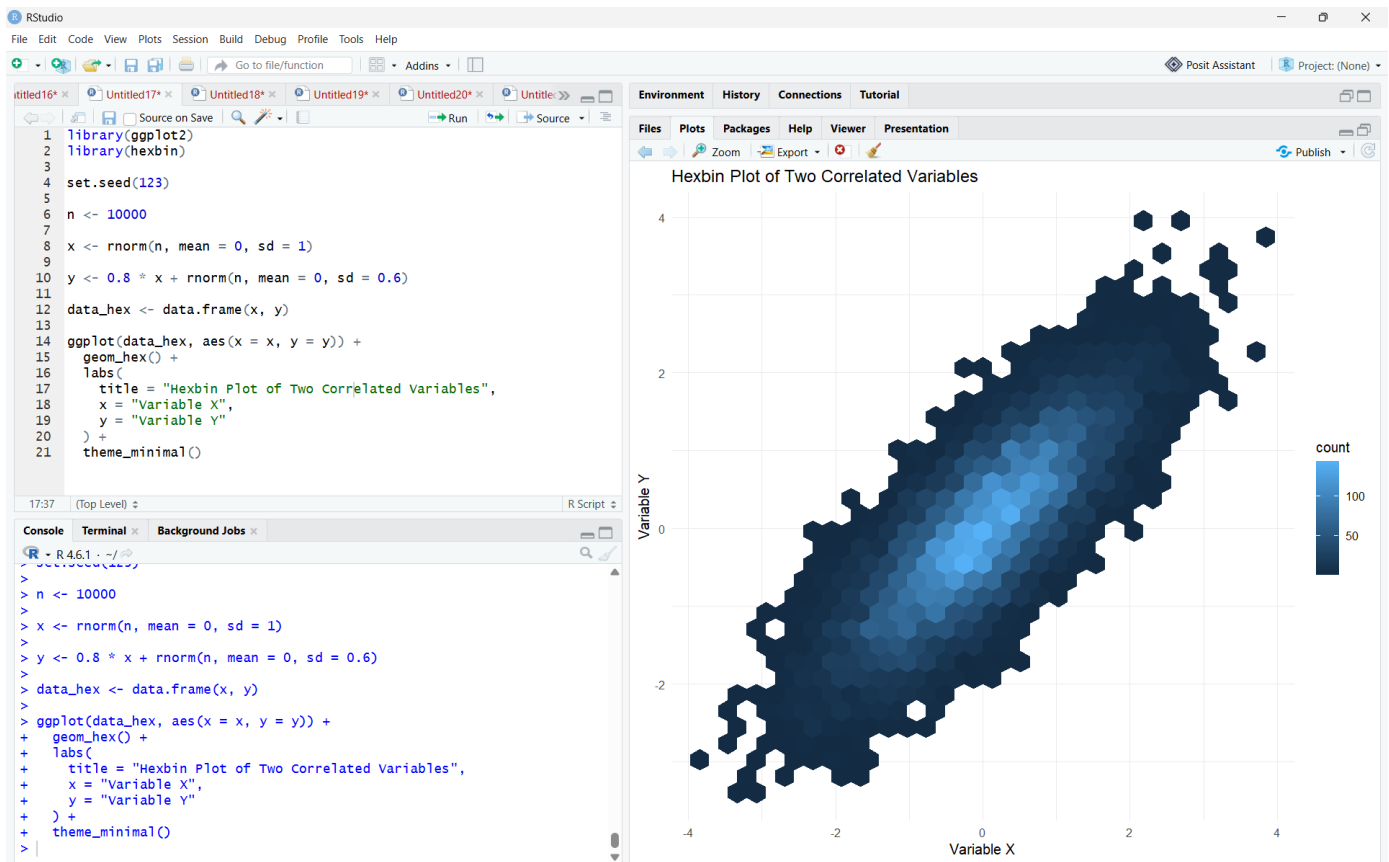
Part a — Hexbin Plot

Dataset provided: Simulated in R (no file needed)

Simulate two correlated variables with 10,000 observations (set.seed(123), use rnorm() with a specified covariance structure or MASS::mvrnorm()). Create a hexbin plot using geom_hex() in ggplot2. Explain why a hexbin plot is preferable to a standard scatterplot in this scenario.

Answer:

Hexbin plots are preferable because they reduce overplotting and clearly show the density distribution of a large number of observations.



Why use a Hexbin Plot?

A standard scatterplot with 10,000 observations can suffer from **overplotting**, where many points overlap and the density of observations becomes difficult to see.

A hexbin plot divides the plotting area into hexagonal bins and represents the number of observations in each bin. Therefore, it makes the **density and concentration of observations much easier to identify**.

Part b — Network Graph

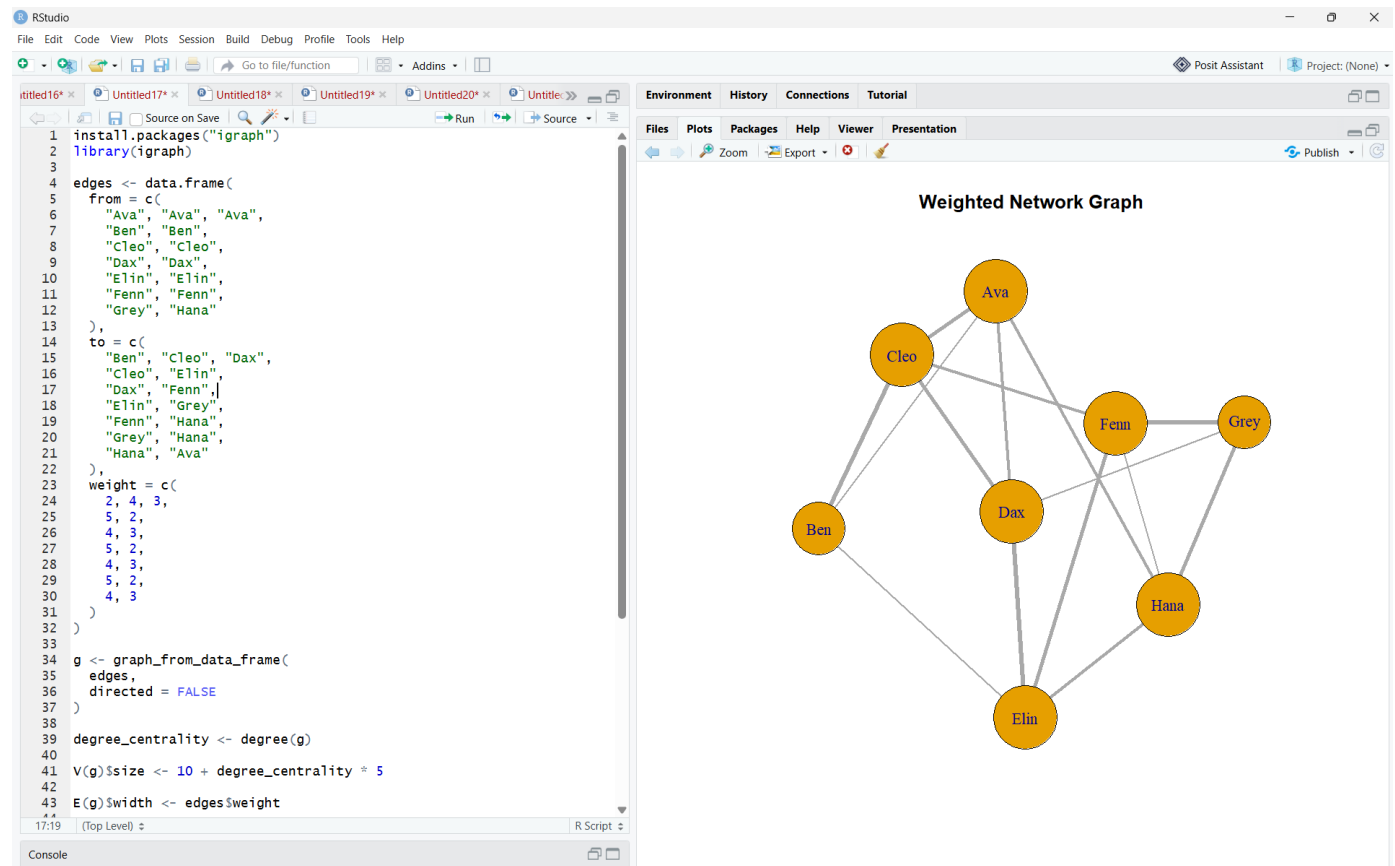
Dataset provided: network_edges.csv

Using the provided edge list (8 nodes, 15 weighted edges), build a network graph using igraph or ggraph. Size nodes by degree centrality and color edges by weight. Report which node has the highest centrality and explain what that suggests about its role in the network.

Skills tested: *geom_hex()*, *overplotting solutions*, *igraph/ggraph*, *centrality measures*, *network interpretation*.

Answer:

The document provides network_edges.csv containing **15 weighted edges and 8 nodes: Ava, Ben, Cleo, Dax, Elin, Fenn, Grey, and Hana**. It asks you to size nodes by degree centrality and color edges by weight.



The document provides network_edges.csv containing **15 weighted edges and 8 nodes: Ava, Ben, Cleo, Dax, Elin, Fenn, Grey, and Hana**. It asks you to size nodes by degree centrality and color edges by weight.